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UND
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PROJEKTARBEIT

Development of the Electronic Actuation & Guidance for Landing and Elevation (EAGLE) system

Entwicklung des Electronic Actuation & Guidance for Landing and
Elevation (EAGLE) Systems

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Abstract

The development of the "Electronic Actuation & Guidance for Landing and Elevation" (EAGLE) system is a project developed in cooperation with the student rocketry association "Spaceflight Rocketry Gießen e.V." as part of its self-developed on-board computer called "Advances System for Control Electronics, Navigation and Tracking" (ASCENT). Therefore, I would like to thank Spaceflight Rocketry Gießen for giving me the opportunity to pursue the project with their equipment and funding.

EAGLE is a hardware-in-the-loop system that uses real-time sensor data to accurately estimate the rockets position and attitude as well as detecting mission critical waypoints (such as apogee) to deploy the recovery system, perform the staging and subsequent firing of the sustainer's motor and make real-time decisions about an active aerodynamical airbreak system, which accurately limits the maximum altitude to a preset value.

The development process of a printed circuit board for this system will be the core subject of this project.

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SPROG		Spaceflight Rocketry Giessen
e.V.		eingetragener Verein
TUDSaT		TU Darmstadt Space Technology
EAGLE		Electronic Acutation & Guidance for Landing and Eleveation
PIPE		Project for Inital Practical Experimentation
ARCHER		Aerial Rocket for Competions and High-altitude Experimental Rocketry
PCB		Printed Circuit Board
PWM		Power Width Modulation
IC		Integrated circuit
PSU		Power supply unit
SMD		Surface mounted devices
SPI		Serial peripheral interface
I2C		Inter-Integrated Circuit
UART		Universal Asynchronous Receiver-Transmitter
RAM		Random access memory
UPDI		Unified Program and Debug Interface
GND		Ground
ASCENT		Advanced System for Control Electronics, Navigation and Tracking
EUROC		European Rocketry Challenge
COTS		Commercial Off-The-Shelf
IMU		Inertial Measurement Unit
JLU		Justus-Liebig-University Giessen
THM		Technische Hochschule Mittelhessen

1 Introduction

"Spaceflight Rocketry Gießen e.V." is a student-founded and ran rocketry association at the Justus Liebig University and University of Applied Sciences in Gießen, which develops and builds experimental rockets, with which the association wants to compete at international student rocketry competitions such as the "European Rocketry Challenge" in Portugal.

1.1 Motivation

In August 2024, SPROG launched its first ever rocket PIPE in Bremen. Due to a lack of testing beforehand, major issues arose regarding the self-developed on-board computer ASCENT during the launch campaign, which were ultimately fixed in time for launch. However, this has sparked a debate about the reliability of our entire launch system, which led to a complete overhaul of all electronic systems whilst also expanding the amount of systems and sensors to increase chances of getting accepted for EuRoC 2026. This meant the introduction of an active aerodynamic airbreak system called EAGLE which was then also fused with the recovery system to save weight and complexity.

1.2 Scope & Requirements

EAGLE sits at the core of SPROG's future rocket "ARCHER" as part of the improved "ASCENT II" computer. ARCHER is a two-staged experimental rocket relying on COTS solid rocket motors and has a diameter of 160 mm. The goal for ARCHER is to compete in EuRoC's 9 km COTS solid category and reach the target altitude as accurately as possible. Due to EuRoC's regulations, EAGLE has to incorporate some system requirements set by the organizers. This means that the initial parachute deployment has to happen "at or near apogee" [1] meaning that EAGLE has to be able to detect apogee, it needs to be able to accurately detect the current height of the rocket to oblige with other rules such as a rule stating: "The main deployment event shall occur at an altitude no higher than 450 m AGL" [1]. EAGLE also needs to be able to actuate servos accurately. Furthermore, it needs to be able to fire pyrotechnic charges on multiple so called "pyro-channels". This means that an igniter needs to be externally connected to the PCB. For safety reasons, these pyro-channels should not be able to fire unless armed through a high signal (3.3 V) coming from an external source. Additionally, the continuity should be measured on these pyro-channels to determine the state of the igniters whilst being vertical on the launch pad. To communicate these igniter states with a ground operator, EAGLE needs to be able to communicate with an external telemetry system. An optional requirement is to be able to program the microcontroller using an USB type C interface. All systems should be SMD as breakout boards take up a lot of

space need to be attached separately by cable leading to extra noise due to vibrations. To achieve this, EAGLE features five sensors including two accelerometers, two pressure sensors and one IMU, which have been decided on based on multiple parameters such as measurement range, resolution, operating voltage and noise level. The microcontroller in charge needs to be able to manage all the data, filter it and make decisions based on the data. In addition to that, there are 4 pyro-channels that allow for a maximum current of 3 A with an external connection terminal and 6 PWM-channels, which will be used for servo actuation. Everything is based on an internally designed PCB layout that fits our current rocket diameter.

2 Selection of components

The components of the PCB make up the most integral part of this system as all decisions and signals are relying on high quality data with low noise. Therefore, a significant amount of time was spent researching a multitude of sensors and other components. As mentioned in 1.2, the board will feature five sensors and a microcontroller powerful enough to handle all the data. The most important requirements for these is an operating voltage of either 3.3 V or 5 V as that is the voltage that is being supplied by the PSU.

2.1 Microcontroller

The microcontroller is the heart of all operations and needs to be the most reliable component on the PCB. It collects all the data from the sensors and distributes it to the telemetry system and data storage as well as fire pyro-channels and actuate the servos and be able to read analogue signals. Therefore, the selected microcontroller needs a wide range of pin-types.

2.1.1 AVR128DB64

The AVR128DB64 was the first choice solely based on past experience and ease of programming. It has an easy to understand pinout and 64 pins and runs on the arduino framework. The AVR128DB28 (same family of controllers) has been used in a previous telemetry system meaning there was a lot of experience using this type of microcontroller. Additionally, it allows communication via SPI, UART and I2C (commonly used communication protocols). The latter being used for communication with all sensors and between the communication with the telemetry subsystem. It also featured a relatively new programming protocol called "UPDI", which only requires one pin for programming instead of two dedicated UART ports like other microcontrollers, making programming substantially easier.

In hindsight, there should have been less emphasis on past experiences as they allowed for an oversight on the computing power of the chip, which is arguable the most important characteristic. After connecting and testing multiple sensors to the microcontroller, it was discovered the RAM overflowed with data during the first loop, rendering the microcontroller useless for usage in EAGLE.

2.1.2 ESP32-S3-Vroom-N4-1u

After running into issues with the AVR128DB64 a team member of the electronics working group within the association suggested to test the ESP32 as it is a dual-core microcontroller and has more RAM available. It also features all common communication protocols, but needs to be programmed using UART, rendering two pins unusable for any other purposes. The programming also needs additional pins called "EN" and "IO0" which need to be pulled

to GND in a specific sequence to put the microcontroller into the so called "programming mode". However, it did feature a D+ and D- pin meaning a USB interface can easily be connected for programming and data-reading. In addition to that, it is an IC meaning that it has additional features such as an built-in WiFi module and antenna. To save space it was decided to use the "-1u" version, because that means that the ESP32 would come without an antenna but with a coax-connector. However, all of this came at the cost of the arduino framework as ESP32s run on their own firmware and framework. In some cases however, the ESP32 is capable of running arduino code, meaning that the code only needs to be adapted slightly.

For those reasons, the ESP32-S3-Vroom-N4-1u was chosen as the microcontroller for EAGLE.

3 References

- [1] Inês d'Ávila. “System Requirement Document”. In: Ref.: PTS_EDU_EuRoC_ST_001928. Mar. 2025. Chap. Recovery Systems and Avionics, pp. 24–35.